

Jamia Millia Islamia

A Central University

DSC-503: COMPUTER NETWORKS

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UNIT 1 : INTRODUCTION

- Introduction to Computer Networks,
- LAN,
- MAN,
- WAN,
- Uses of Computer Networks,
- LAN Technologies- Transmission Topologies, Access methods.
- Network Architecture, Protocol and standards,
- References Model: OSI-ISO and TCP/IP,
- Overview: Circuit switching, Message switching and Packet switching.

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DATA COMMUNICATION

- **Data**
  - refers to the raw facts that are collected while
- **Information**
  - refers to processed data that enables us to take decisions.
- **Data Communication**
  - is a process of exchanging data or information
- In case of computer networks this exchange is done between two devices over a transmission medium.

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Characteristics of Data Communication

- The effectiveness of any data communications system depends upon the following four fundamental characteristics:
  1. **Delivery**:
     - The data should be delivered to the correct destination and correct user.
  2. **Accuracy**:
     - The communication system should deliver the data accurately, without introducing any errors.

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Characteristics of Data Communication

3. **Timeliness:**
   - Audio and Video data has to be delivered in a timely manner without any delay.

4. **Jitter:**
   - It is the variation in the packet arrival time.

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Components of Data Communication

- A Data Communication system has five components as shown in the diagram below:

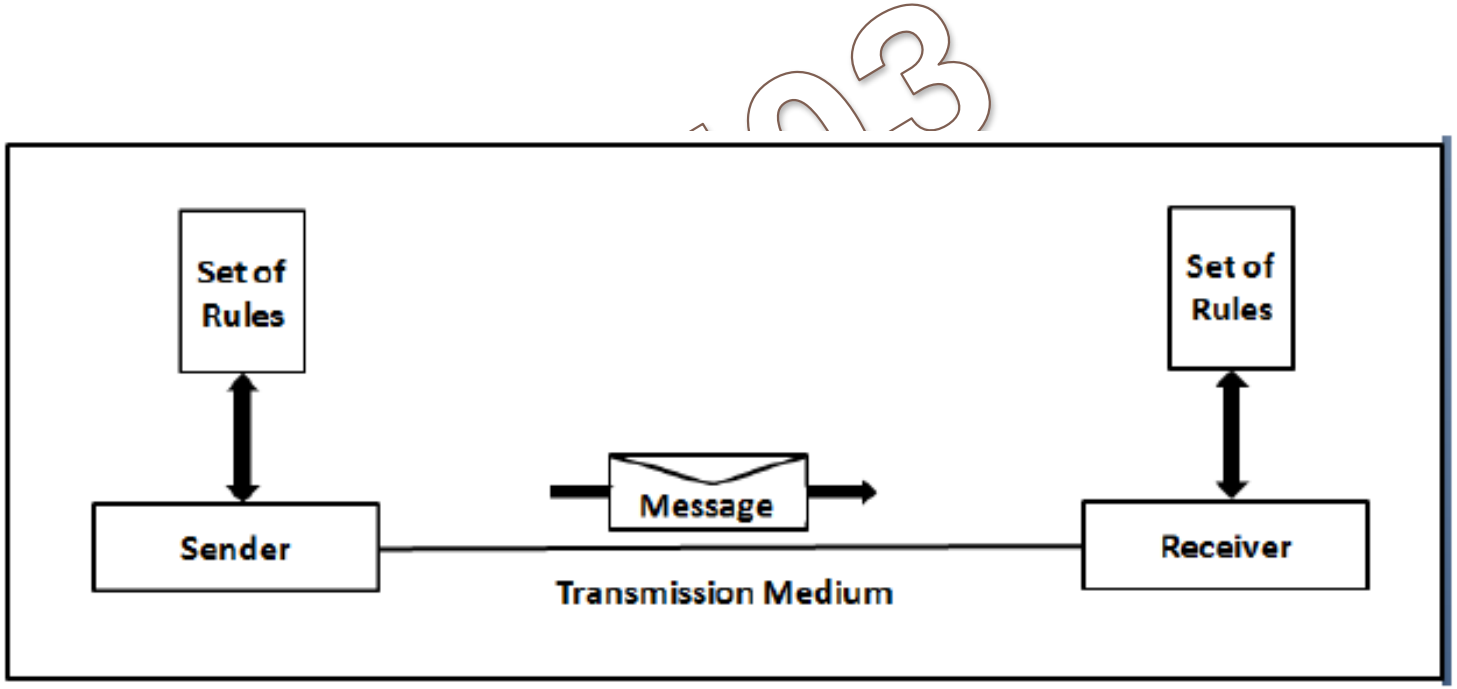


Image adapted From: *"Forouzan, A. Behrouz. Data Communications and Networking"*

Components of Data Communication

1. **Sender :**
   - The sender is any device that is capable of sending the data (message).
2. **Receiver :**
   - The receiver is a device that the sender wants to communicate the data (message).
3. **Message :**
   - Message is the information to be communicated by the sender

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Components of Data Communication

4. **Transmission Medium:**

- It is the path by which the message travels from sender to receiver.
- It can be wired or wireless and many subtypes in both. to the receiver.

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Components of Data Communication: Protocol

5. **Protocol :**
   - It is an agreed upon set or rules used by the sender and receiver to communicate data.
   - A protocol is a set of rules that governs data communication.

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Elements of a Protocol

- **Elements of a Protocol :**
  - **I. Syntax :**
    - It means the structure or format of the data.
    - It is the arrangement of data in a particular order
  - **II. Semantics**
    - It tells the meaning of each section of bits and indicates the interpretation of each section.
  - **III. Timing**
    - It tells the sender about the readiness of the receiver to receive the data

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DATA REPRESENTATION

- Data is collection of raw facts which is processed to deduce information.
- There may be different forms in which data may be represented.

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Forms of Data Representation

- **Text :**
  - Text includes combination of alphabets in small case as well as upper case.
  - It is stored as a pattern of bits.
  - Prevalent encoding system : ASCII, Unicode
- **Numbers :**
  - Numbers include combination of digits from 0 to 9.
  - It is stored as a pattern of bits.

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Forms of Data Representation ...

- **Images:**
  - A Pixel is the smallest element of an image.
  - To put it in simple terms, a picture or image is a matrix of pixel elements
  - The pixels are represented in the form of bits.
  - The size of an image depends upon the number of pixels
  - Commonly used Image formats : jpg, png, bmp, etc

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Forms of Data Representation ...

- **Audio :**
  - Data can also be in the form of sound which can be recorded and broadcasted.
  - Audio data is continuous, not discrete.

- **Video :**
  - Video refers to broadcasting of data in form of picture or movie

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COMPUTER NETWORK

- Computer Networks are used for data communications
  - A computer network can be defined as a collection of nodes.
  - A node can be any device capable of transmitting or receiving data.
  -
  - The communicating nodes have to be connected by communication links.

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COMPUTER NETWORK

- A Compute network should ensure
  - **reliability**
  - **security**
  - **performance**

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INTRODUCTION

- **Communication** is the process of establishing connection or link between two points for information exchange.

OR

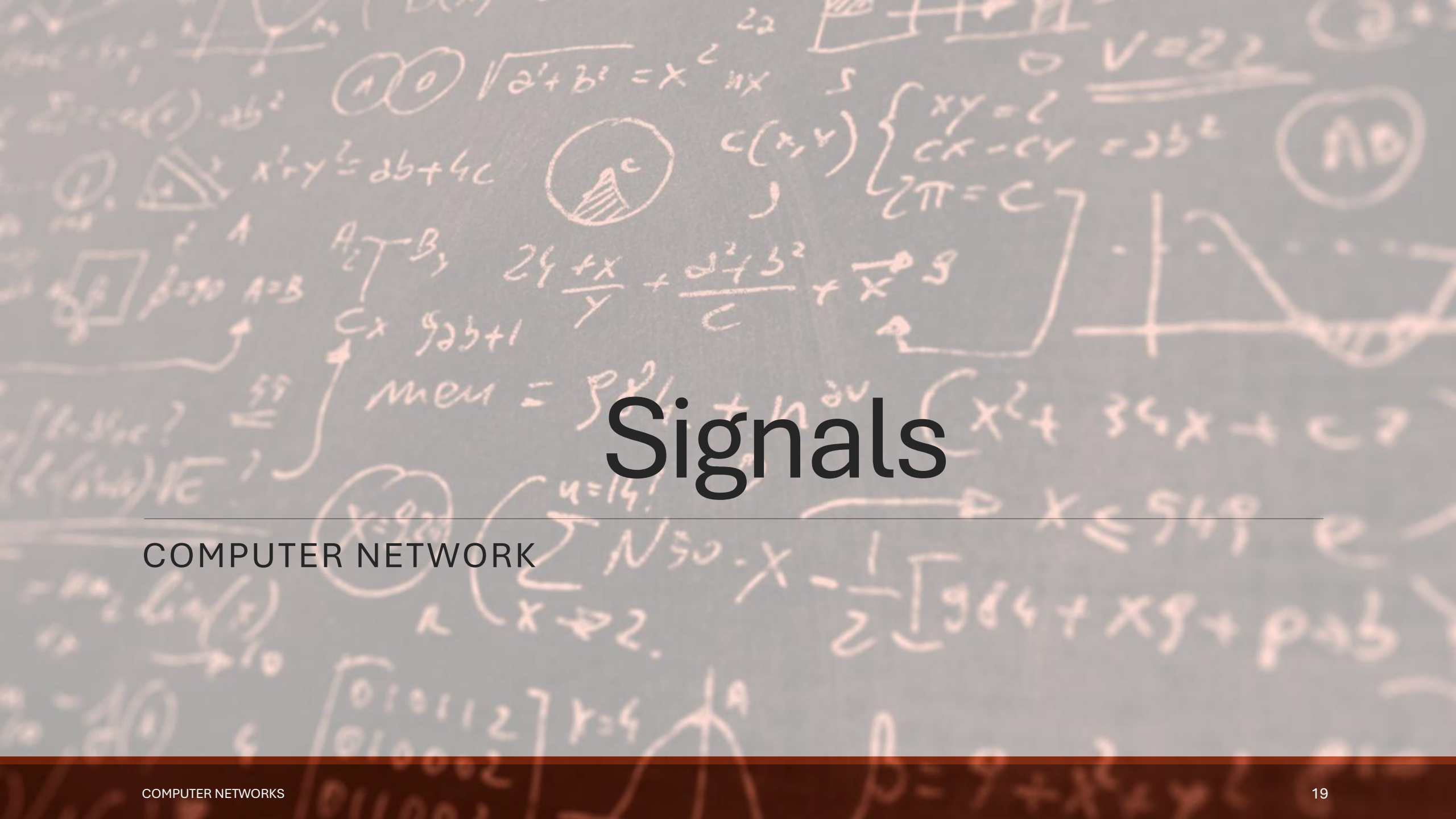
- **Communication** is simply the basic process of exchanging information.
- The electronic equipment which are used for communication purpose, are called **communication equipment**.
- Different communication equipment when assembled together form a **communication system**.

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SOURCES OF INFORMATION

- Some of the important sources of information in the communication environment may be listed as under :
  1. Speech
  2. Music
  3. Pictures
  4. Computer data
- A source of information is basically a signal which carries the information.

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Signals

COMPUTER NETWORK

Signal

- **A signal** may be defined as the single valued function of time.
  - Time plays the role of an independent variable.
  - This means that at every instant of time, the signal has a unique value.
- **Signal** is a time varying physical phenomenon which is intended to convey information.

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Classifications of Signals

- Signals are classified into the following categories:

1. Continuous Time and Discrete Time Signals
2. Analog and Digital Signals
3. Periodic and Aperiodic Signals
4. Finite vs. Infinite Length Signals

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Classifications of Signals

5. Causal vs. Anticausal vs. Noncausal Signals
6. Even and Odd Signals
7. Deterministic and Random Signal

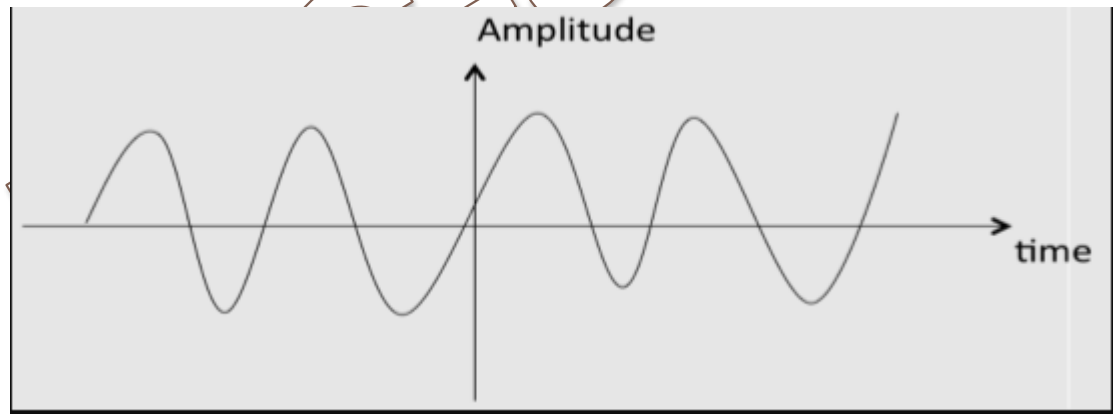
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1. Continuous Time and Discrete Time Signals

- A signal is said to be continuous when it is defined for all instants of time.

Signal →f = f(t)

t is independent variable



Continuous Time and Discrete Time Signals

- A signal is said to be discrete when it is defined at only discrete instants of time.

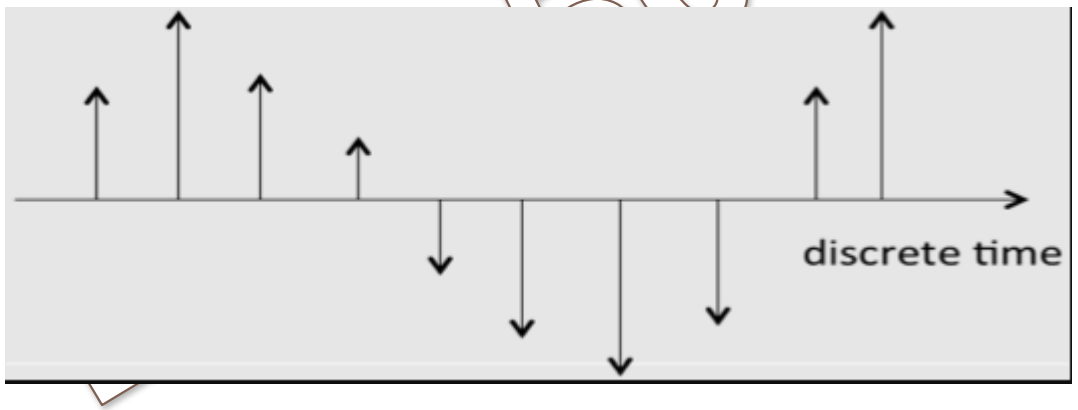


Image adapted From: *"Sharma, Sanjay. Data Communications "*

2. Analog vs. Digital

- The difference between analog and digital is similar to the difference between continuous-time and discrete-time.
- Here the difference involves the values of the function.
- Analog corresponds to a continuous set of possible function values,
- Digital corresponds to a discrete set of possible function values.
- Example:
  - Digital signal - Binary sequence:
    - where the values of the function can only be one or zero.

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Analog Signal

- Continuous time analog

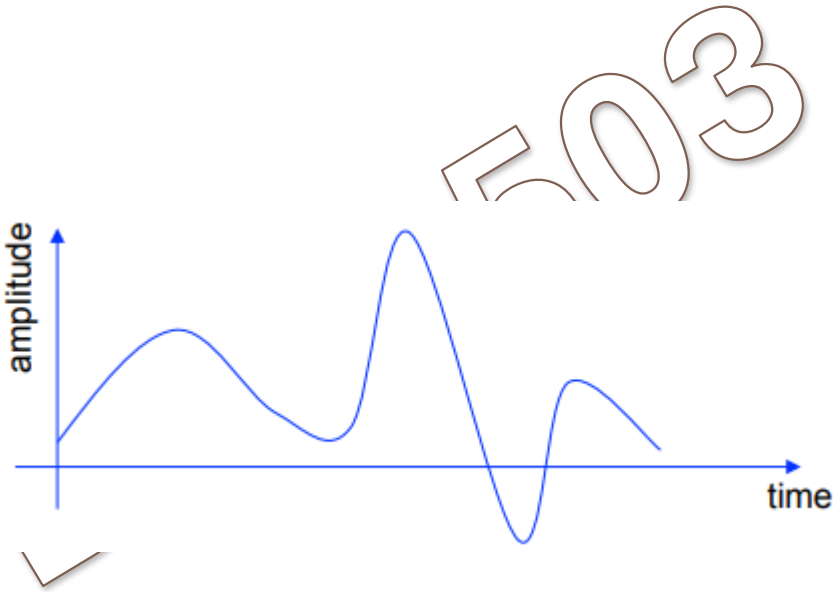


Image adapted From: *"Sharma, Sanjay. Data Communications "*

Analog vs. Digital

- Continuous time digital (or quantized)

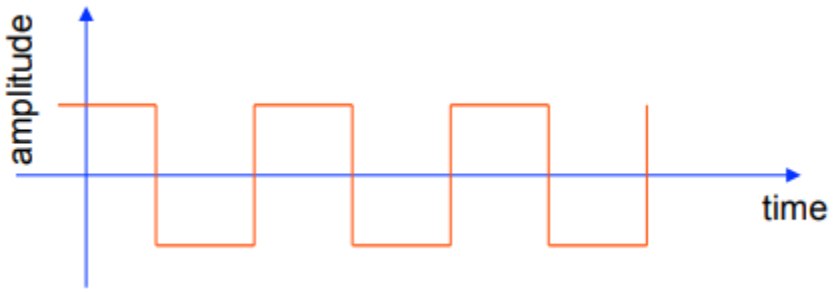


Image adapted From: *"Sharma, Sanjay. Data Communications "*

Analog vs. Digital

- Discrete time digital

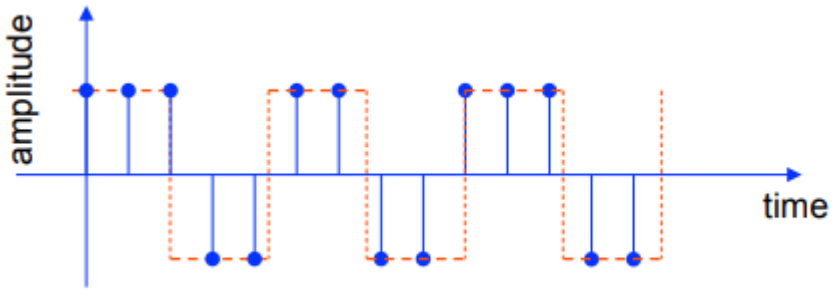


Image adapted From: *"Sharma, Sanjay. Data Communications "*

3. Periodic vs. Aperiodic

- Periodic signals repeat with some period T, while aperiodic, or non-periodic, signals do not.
- An aperiodic function never repeats, although technically an aperiodic function can be considered like a periodic function with an infinite period.
- We can define a periodic function through the following mathematical expression, where t can be any number and T is a positive constant:

$$f(t) = f(T + t)$$

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Periodic vs. Aperiodic

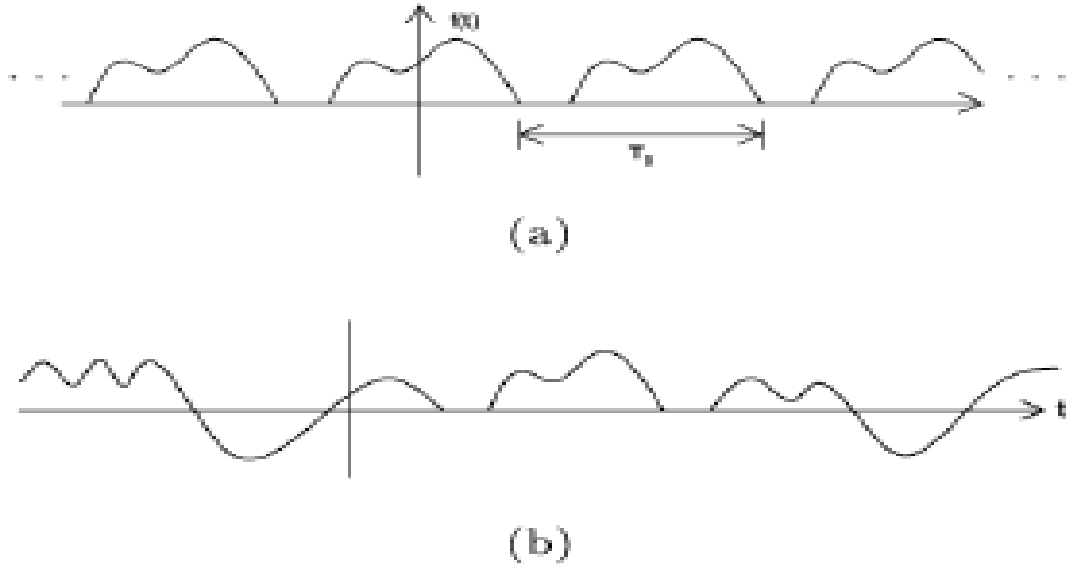


Figure : (a) A periodic signal with period T_0
(b) An aperiodic signal

Image adapted From: *"Sharma, Sanjay. Data Communications "*

4. Finite vs. Infinite Length

- Signals can be characterized as to whether they have a finite or infinite length set of values.
- Most finite length signals are used when dealing with discrete-time signals or a given sequence of values.
- Mathematically,
- f (t) is a finite-length signal if it is nonzero over a finite interval

$$t1 < f(t) < t2$$

where t1 > 1 and t2 < 1.

- Similarly, an infinite-length signal, f (t), is nonzero over all real numbers:

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5. Causal vs. Anticausal vs. Noncausal

- Causal signals are signals that are zero for all negative time.

F(t) =0 t <0

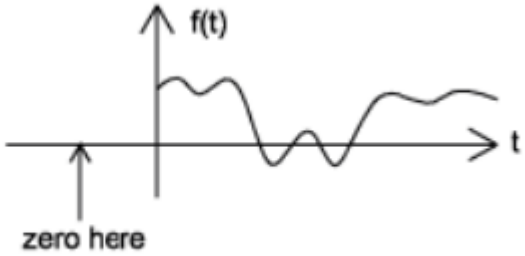


Image adapted From: *"Sharma, Sanjay. Data Communications "*

Causal vs. Anticausal vs. Noncausal

- Anti-causal are signals that are zero for all positive time.

F(t) =0 t >=0

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Image adapted From: *"Sharma, Sanjay. Data Communications "*

Causal vs. Anticausal vs. Noncausal

- Non-causal signals are signals that have non-zero values in both positive and negative time.

$\exists t_1 < 0:$ $f(t_1) = 0$

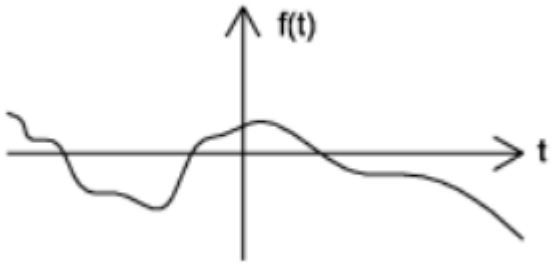


Image adapted From: *"Sharma, Sanjay. Data Communications "*

6. Even vs. Odd

- An even signal is any signal f such that

$$f(t) = f(-t).$$

- Even signals can be easily spotted as they are symmetric around the vertical axis.

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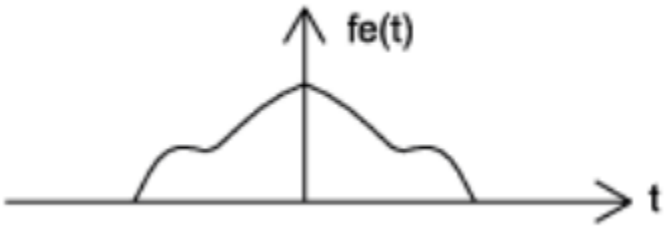


Image adapted From: *"Sharma, Sanjay. Data Communications "*

6. Odd Signal

- An odd signal, on the other hand, is a signal f such that

$$f(t) = -(f(-t))$$

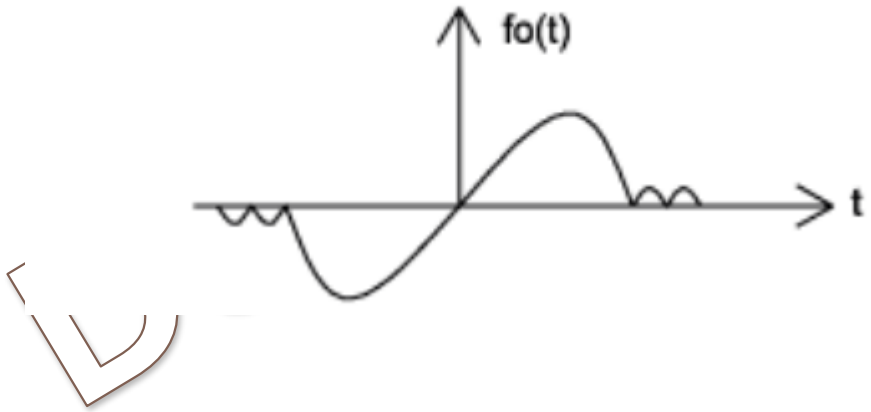


Image adapted From: *"Sharma, Sanjay. Data Communications "*

7. Deterministic and Non-deterministic Signals

- A signal is said to be deterministic if there is no uncertainty with respect to its value at any instant of time.
- Signals which can be defined exactly by a mathematical formula are known as deterministic signals.

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Image adapted From: *"Sharma, Sanjay. Data Communications "*

Deterministic and Non-deterministic Signals

- A signal is said to be non-deterministic/Random/Probablistic, if there is uncertainty with respect to its value at some instant of time.

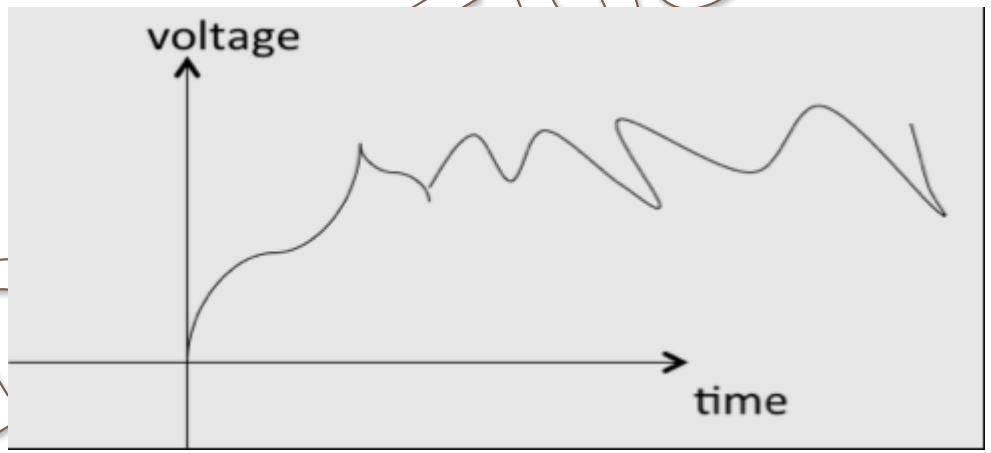


Image adapted From: *"Sharma, Sanjay. Data Communications "*

Deterministic and Non-deterministic Signals

- Non-deterministic signals are random in nature hence they are called random signals.
- Random signals cannot be described by a mathematical equation.
- They are modelled in probabilistic terms.

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Refer Class Notes and Recommended Books for further Reading

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Reference Books

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Thank You

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